

Configuring sFlow

This chapter describes how to configure sFlow on Cisco NX-OS devices.

This chapter includes the following sections:

- sFlow, on page 1
- Prerequisites for sFlow, on page 2
- Guidelines and Limitations for sFlow, on page 2
- Default Settings for sFlow, on page 4
- Configuring sFlow , on page 4
- Verifying the sFlow Configuration, on page 11
- Monitoring and Clearing sFlow Statistics, on page 11
- Configuration Examples for sFlow, on page 12
- Additional References, on page 12

sFlow

Sampled flow (sFlow) is a sampled flow monitoring technology that:

- enables you to monitor real-time traffic in data networks with switches and routers,
- employs the sampling mechanism in sFlow agent software to monitor traffic, and
- forwards sample data to a central data collector.

For more information about sFlow, see RFC 3176.

sFlow Agent

The sFlow agent, which is embedded in the Cisco NX-OS software, periodically samples or polls the interface counters that are associated with a data source of the sampled packets. The data source can be an Ethernet interface, an EtherChannel interface, or a range of Ethernet interfaces. The sFlow agent queries the Ethernet port manager for the respective EtherChannel membership information and also receives notifications from the Ethernet port manager for membership changes.

When you enable sFlow sampling, based on the sampling rate and the hardware internal random number, the ingress packets and egress packets are sent to the CPU as an sFlow-sampled packet. The sFlow agent processes the sampled packets and sends an sFlow datagram to the sFlow analyzer. In addition to the original sampled

packet, an sFlow datagram includes information about the ingress port, the egress port, and the original packet length. An sFlow datagram can have multiple sFlow samples.

Prerequisites for sFlow

sFlow has the following prerequisites:

- For Cisco Nexus 9332PQ, 9372PX, 9372TX, and 93120TX switches and for Cisco Nexus 9396PX, 9396TX, and 93128TX switches with the N9K-M6PQ generic expansion module (GEM), you must configure the sFlow and SPAN ACL TCAM region sizes for any uplink ports that are to be configured as an sFlow data source. To do so, use the **hardware access-list tcam region sflow** and **hardware access-list tcam region span** commands. See Configuring ACL TCAM Region Sizes for more information.



Note By default, the sflow region size is zero, and the span region size is non-zero. You need to configure the sflow region to 256 and allocate enough entries to the span region in order to configure the port as an sFlow data source.

- Egress sFlow of multicast traffic requires **hardware multicast global-tx-span** configuration

Guidelines and Limitations for sFlow



Note For scale information, see the release-specific *Cisco Nexus 9000 Series NX-OS Verified Scalability Guide*.

sFlow has the following guidelines and limitations:

- If at least one sFlow data source is configured, the SPAN sessions cannot be brought up.
  - If at least one SPAN session is configured as **no shut** , sFlow data sources cannot be added.
  - The sampling mode that is used for sFlow is based on an algorithm that is known as LFSR. Due to the use of LFSR, it is not guaranteed that one in every few packets are sampled with the sampling rate of n. However, the number of packets that are sampled is equal to the total packets over a period of time.
- When sFlow is used to sample the Rx traffic from FEX HIF ports, additional VNTAG and 802.1q tags are present in the sampled traffic.
- In Cisco Nexus 9300-EX and 9300-FX platform switches, the FEX, HIF, and NIF ports cannot be configured as sFlow data-source interfaces.
- When sFlow and SPAN are configured on the same interface, and the hardware rate-limiter is configured for sFlow, the Rate-Limiter Drops counter in the output of the **show hardware rate-limiter** command displays more drops than expected.

- sFlow is a software-driven feature, hardware only sends copies of traffic from the sFlow source interfaces to the CPU for further processing. Elevated CPU usage is expected. sFlow traffic sent to the CPU by hardware is rate-limited to protect the CPU.
- When you enable sFlow for an interface, it is enabled for both ingress and egress. You cannot enable sFlow for only ingress or only egress.

  For Cisco Nexus 9508 switches with Cisco Nexus 9636C-R and 9636Q-R line cards, sFlow can be enabled for an interface only in the ingress direction.
- The storm control feature does not work if you enable storm control on an interface where sFlow is also enabled.
- sFlow is not supported on the SVIs.
- Subinterfaces are not supported for sFlow.
- We recommend you configure the sampling rate that is based on the sFlow configuration and traffic in the system.
- The switch supports only one sFlow collector.
- sFlow and Network Address Translation (NAT) are not supported on the same port.
- sFlow supports sampling IPv6 traffic but only on IPv4 collector address.
- sFlow does not support egress sampling for multicast, broadcast, or unknown unicast packets.
- sFlow counters increment even for control packets that ingress on the sFlow data-source interfaces. These packets may be sampled and send out as sFlow datagrams (similar to data plane traffic).
- The following Cisco Nexus switches support sFlow and SPAN together:
  - N9336C-FX2
  - N93240YC-FX2
  - N93360YC-FX2
- Beginning with Cisco NX-OS Release 9.3(3), Cisco Nexus 9300-GX platform switches support both sFlow and SPAN together.
- Nexus 9000-EX, FX, GX family of switches only support sampling at the following values: 4096, 8192, 16384, 32768, 65536. Configuring values other than these results in the value being rounded off to the next supported value.
- When sFlow is configured on N9K-C9508-FM-G with the N9K-X9716D-GX line card, disable sFlow before configuring SPAN sessions.
- Beginning with Cisco NX-OS Release 10.1(1), sFlow and SPAN are supported on Cisco N9K-C93180YC-FX3 platform switches.
- Beginning with Cisco NX-OS Release 10.1(2), sFlow is supported on the Cisco N9K-X9624D-R2 line card.
- Beginning with Cisco NX-OS Release 10.1(2), sFlow supports VXLAN traffic on the Cisco N9K-C9508-FM-G cloud-scale fabric module with the N9K-X9716D-GX line card.

- Beginning with Cisco NX-OS Release 10.2(1), extended BGP data can now be collected. In order for sFlow to collect this data, a non-SVI Layer 3 interface such as a physical interface or port-channel must be configured as the sFlow source.

Default Settings for sFlow

The following table lists the default settings for sFlow parameters.

Table 1: Default sFlow Parameters

Parameters	Default
sFlow sampling rate	4096
sFlow sampling size	128
sFlow counter poll interval	20
sFlow maximum datagram size	1400
sFlow collector IP address	0.0.0.0
sFlow collector port	6343
sFlow agent IP address	0.0.0.0

Configuring sFlow

Enabling sFlow

You must enable the sFlow feature before you can configure sFlow settings on the switch.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **feature sflow** **Example:** `switch(config)# feature sflow`	Enables or disables sFlow.
Step 3	(Optional) **show feature** **Example:**	Displays the enabled and disabled features.

	Command or Action	Purpose
	`switch(config)# show feature`	
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the Sampling Rate

You can configure the sampling rate for sFlow.

Before you begin

Make sure that you have enabled sFlow.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **sflow sampling-rate** *sampling-rate* **Example:** `switch(config)# sflow sampling-rate 50000`	Configures the sFlow sampling rate for packets. The *sampling-rate* can be an integer between 4096 and 1000000000. **Note** Nexus 9000-EX, FX, GX, FX2, FX3, family of switches support sampling only at the following values: 4096, 8192, 16384, 32768, 65536. Configuring values other than these will result in the value being rounded off to the next supported value.
Step 3	(Optional) **show sflow** **Example:** `switch(config)# show sflow`	Displays the sFlow configuration.
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the Maximum Sampled Size

You can configure the maximum number of bytes that should be copied from a sampled packet.

Before you begin

Make sure that you have enabled sFlow.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **sflow max-sampled-size** *sampling-size* **Example:** `switch(config)# sflow max-sampled-size 200`	Configures the sFlow maximum sampling size. The range for the *sampling-size* is from 64 to 256 bytes.
Step 3	(Optional) **show sflow** **Example:** `switch(config)# show sflow`	Displays the sFlow configuration.
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the Counter Poll Interval

You can configure the maximum number of seconds between successive samples of the counters that are associated with the data source. A sampling interval of 0 disables counter sampling.

Before you begin

Make sure that you have enabled sFlow.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

	Command or Action	Purpose
Step 2	[**no**] **sflow counter-poll-interval** *poll-interval* **Example:** `switch(config)# sflow counter-poll-interval 100`	Configures the sFlow poll interval for an interface. The range for the *poll-interval* is from 0 to 2147483647 seconds.
Step 3	(Optional) **show sflow** **Example:** `switch(config)# show sflow`	Displays the sFlow configuration.
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the Maximum Datagram Size

You can configure the maximum number of data bytes that can be sent in a single sample datagram.

Before you begin

Make sure that you have enabled sFlow.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **sflow max-datagram-size** *datagram-size* **Example:** `switch(config)# sflow max-datagram-size 2000`	Configures the sFlow maximum datagram size. The range for the *datagram-size* is from 200 to 9000 bytes.
Step 3	(Optional) **show sflow** **Example:** `switch(config)# show sflow`	Displays the sFlow configuration.
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the sFlow Collector Address

You can configure the IPv4 address of the sFlow data collector that is connected to the management port.

Before you begin

Make sure that you have enabled sFlow.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **sflow collector-ip** *ip-address* **vrf** *vrf* [**source** *ip-address*] **Example:** `switch(config)# sflow collector-ip 192.0.2.5 vrf management`	Configures the IPv4 address for the sFlow collector. If the IP address is set to 0.0.0.0, all samples will be dropped. The *vrf* can be one of the following: • A user-defined VRF name—You can specify a maximum of 32 alphanumeric characters. • **vrf management**—You must use this option if the sFlow data collector is on the network connected to the management port. • **vrf default**—You must use this option if the sFlow data collector is on the network connected to the front-panel ports. The **source** *ip-address* option causes the sent sFlow datagram to use the source IP address as the IP packet source address. The source IP address has to be already configured on one of the switch local interfaces; otherwise, an error message appears. If the interface with the source IP address is changed or removed after this option is configured, the sFlow datagram will no longer be sent out, and an event history error and syslog error will be logged. When the **source** *ip-address* option is not configured, Cisco NX-OS picks the IP packet source address automatically for the sent sFlow datagram.
Step 3	(Optional) **show sflow**	Displays the sFlow configuration.

	Command or Action	Purpose
	Example: `switch(config)# show sflow`	
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the sFlow Collector Port

You can configure the destination port for sFlow datagrams.

Before you begin

Make sure that you have enabled sFlow.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **sflow collector-port** *collector-port* **Example:** `switch(config)# sflow collector-port 7000`	Configures the UDP port of the sFlow collector. The range for the *collector-port* is from 1 to 65535.
Step 3	(Optional) **show sflow** **Example:** `switch(config)# show sflow`	Displays the sFlow configuration.
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the sFlow Agent Address

You can configure the IPv4 address of the sFlow agent.

Before you begin

Make sure that you have enabled sFlow.

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **sflow agent-ip** *ip-address* **Example:** `switch(config)# sflow agent-ip 192.0.2.3`	Configures the IPv4 address of the sFlow agent. The default IP address is 0.0.0.0, which means that all samples will be dropped. You must specify a valid IP address to enable sFlow functionality. **Note** This IP address is not necessarily the source IP address for sending the sFlow datagram to the collector.
Step 3	(Optional) **show sflow** **Example:** `switch(config)# show sflow`	Displays the sFlow configuration.
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Configuring the sFlow Sampling Data Source

You can configure the source of the data for the sFlow sampler as an Ethernet port, a range of Ethernet ports, or a port channel.

Before you begin

Make sure that you have enabled sFlow.

If you want to use a port channel as the data source, make sure that you have already configured the port channel and you know the port channel number.

Make sure that the sFlow and SPAN ACL TCAM region sizes are configured for any uplink ports that are to be configured as an sFlow data source on the following devices: Cisco Nexus 9332PQ, 9372PX, 9372TX, and 93120TX switches and Cisco Nexus 9396PX, 9396TX, and 93128TX switches with the N9K-M6PQ generic expansion module (GEM).

Procedure

	Command or Action	Purpose

Step 1	**configure terminal**
	Example:
	`switch# configure terminal`
	`switch(config)#`
		Enters global configuration mode.

Step 2	[**no**] **sflow data-source interface** [**ethernet** *slot*/*port*[*-port*] | **port-channel** *channel-number*] **Example:** `switch(config)# sflow data-source interface ethernet 1/5-12`	Configures the sFlow sampling data source. For an Ethernet data source, *slot* is the slot number, and *port* can be either a single port number or a range of ports designated as *port-port*.
Step 3	(Optional) **show sflow** **Example:** `switch(config)# show sflow`	Displays the sFlow configuration.
Step 4	(Optional) **copy running-config startup-config** **Example:** `switch(config)# copy running-config startup-config`	Copies the running configuration to the startup configuration.

Verifying the sFlow Configuration

Use these commands to display the sFlow configuration.

Table 2: sFlow Show Commands

Command	Purpose
show sflow	Displays all the data sources of the sFlow samplers and the sFlow agent configuration.
show process	Verifies whether the sFlow process is running.
show running-config sflow [**all**]	Displays the current sFlow running configuration.

Monitoring and Clearing sFlow Statistics

Use the **show sflow statistics** command to display the sFlow statistics.

Use the following commands to clear the sFlow statistics:

Command	Description
clear sflow statistics	Clears most of the sFlow statistics from the **show sflow statistics** command.
clear counters interface all	Clears the Total Packets field from the **show sflow statistics** command.
clear hardware rate-limiter sflow	Clears the Total Samples field from the **show sflow statistics** command.

Configuration Examples for sFlow

This example shows how to configure sFlow:

```
feature sflow
sflow sampling-rate 4096
sflow max-sampled-size 200
sflow counter-poll-interval 100
sflow max-datagram-size 2000
sflow collector-ip 192.0.2.5 vrf management
sflow collector-port 7000
sflow agent-ip 192.0.2.3
sflow data-source interface ethernet 1/5
```

Additional References

Related Documents

Related Topic	Document Title
ACL TCAM regions	Configuring IP ACLs